

Central Limit Theorem Notes

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Introduction

The Central Limit Theorem (CLT) is one of the most important results of probability theory. It states that the sample mean of a large number of independent identically distributed (i.i.d.) random variables with finite mean and variance tends toward the standard normal distribution, irrespective of the original distribution.

Formally, if $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$, then

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \Phi(x),$$

where $\Phi(x)$ is the cumulative distribution function (CDF) of the standard normal distribution $N(0, 1)$.

Central Limit Theorem (Moment Generating Function Proof)

Proof

Let $X_1, X_2, \dots$ be i.i.d. random variables with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 \in (0, \infty).$$

Define

$$Z_n = \frac{1}{\sigma\sqrt{n}} \sum_{i=1}^n (X_i - \mu).$$

If the moment generating function (MGF) of X_1 exists in a neighborhood of 0, then

$$Z_n \xrightarrow{d} N(0, 1).$$

Calculation

Define

$$Y_i = \frac{X_i - \mu}{\sigma}.$$

Then

$$\mathbb{E}[Y_i] = 0, \quad \text{Var}(Y_i) = 1.$$

Thus,

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i.$$

The MGF of a random variable W is defined as

$$M_W(t) = \mathbb{E}[e^{tW}], \quad t \in \mathbb{R}.$$

Since Y_i are i.i.d., the MGF of Z_n is

$$M_{Z_n}(t) = \left(M_Y\left(\frac{t}{\sqrt{n}}\right) \right)^n.$$

Step 1: Expansion of $M_Y(u)$

By Taylor expansion around $u = 0$ (using $\mathbb{E}[Y_i] = 0, \text{Var}(Y_i) = 1$):

$$M_Y(u) = 1 + \frac{u^2}{2} + o(u^2), \quad u \rightarrow 0.$$

Step 2: Substitute $u = \frac{t}{\sqrt{n}}$

$$M_Y\left(\frac{t}{\sqrt{n}}\right) = 1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right).$$

Step 3: Compute $M_{Z_n}(t)$

$$M_{Z_n}(t) = \left(1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n.$$

Taking logarithms,

$$\ln M_{Z_n}(t) = n \ln \left(1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right).$$

Using $\ln(1+x) = x - \frac{x^2}{2} + o(x^2)$ as $x \rightarrow 0$:

$$\ln M_{Z_n}(t) = n \left(\frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right).$$

Hence,

$$\ln M_{Z_n}(t) \rightarrow \frac{t^2}{2}, \quad \text{as } n \rightarrow \infty.$$

Exponentiating,

$$M_{Z_n}(t) \rightarrow e^{t^2/2}.$$

Step 4: Conclusion

The function

$$M(t) = e^{t^2/2}$$

is the MGF of the standard normal distribution $N(0, 1)$.

By Curtiss's theorem (convergence of MGFs in a neighborhood of 0 implies convergence in distribution),

$$Z_n \xrightarrow{d} N(0, 1).$$

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